MILITARY SPECIFICATION

REFRIGERATOR-CONTAINER, FIELD, 8 FEET BY 8 FEET BY 10 FEET

This specification is approved for use by Natick Research and Development Command, Department of the Army, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an 3 foot by 8 foot by 10 foot insulated container for use with a 4,000 British Thermal Unit (B.t.u.) per hour, electric motor- driven refrigeration unit. The refrigeration unit is not to be furnished or installed by the container supplier.

2. APPLICABLE D0CUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-S-896/1	Switches, toggle (Toggle and Lock) , Flush Mlounted (AC-DC 'T' Rated) (Detail Specification)
NN-P-530	Plywood, Flat Panel
QQ-A-250/8	Aluminum Alloy 5052, Plate and Sheet
TT-C-490	Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-C-520	Coating Compound, Bituminous, Solvent Type, Underbody, (For Motor Vehicle)
TT-P-1757	primer Coating, Zinc Chromate, Low Moisture Sensitivity
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-60	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any Pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760. by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this

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MIL-T-704 - Treatment and painting of Materiel
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-10578 - Corrosion Removing and Metal Conditioning Compound (phosphoric Acid Base)
MIL-S-40144 - Shelving, Storage, Stationary and Mobile, Food Service
MIL-E-52798 - Enamel, Alkyd, Camouflage

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MS 35647 - Padlock, Key operated

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-2853 - Unit Opening, Container, Insulated, 8'x8'x10'
5-13-2854 - Container, Insulated, 8'x8'x10'

A miniature copy of 5-13-2854 identified as Figure 1 is attached for information purposes only.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in " effect on date of invitation for bids or request for proposal shall apply:

LAWS AND REGULATIONS

NATIONAL BUREAU OF STANDARDS

Handbook H 28 - Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-1692 - Rate of Burning or Extent of Burning of Cellular Plastics Using. a Specimen Supported by a Horizontal Screen

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN NATIONAL STANDARDS INSTITUTE

MH 5.1.1M - Requirements for Closed Van Cargo Containers

(Application for copies should be addressed to The American Society of Mechanical Engineers, United Engineering Center/345E. 47th Street, New York,NY 10017 .)

Title 49, Code of Federal Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Underwriters Laboratories, Inc. (UL)

(Application for copies should be addressed to Underwriters Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611.

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

3. REQUIREMENTS

3,1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish two complete containers for first article inspection and approval (see 4.3 and 6.2).

3.2 Safety standard, All electrical components shall conform to the UL standards for safety, as applicable.

3.3 Materials and components. Materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer in his commercial containers, provided the completed item complies with all provisions of this specification (see 6.5).

3.3.1 Fastening devices. Steel bolts, nuts, screws, and washers shall be zinc or plated. Threads shall conform to the National Bureau of Standards Handbook H-28.

3.3.2 Insulation material. The insulation shall be polyurethane and shall have a manufacturer's commercial rated "K" factor of not more than 0.15 Btu per hour per square foot per inch of thickness per degree Fahrenheit (F.) temperature difference at a mean temperature of 75°F. The insulation shall be fire retardent and self-extinguishing as determined by ASTM D-1692. To meet the relative burning characteristics the words "Fire retardent and self-extinguishing" shall be construed to mean that the insulation shall extinguish within 60 seconds after removal from the burner.

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3.3.3 Metal conditioners.

3.3.3.1 Ferrous metal. Ferrous metal conditioner shall be type I or II of MIL-C-10578 or shall conform to type I or II method VI of TT-C-490.

3.3.3.2 Aluminum. Aluminum surface conditioner shall conform to MIL-C-5541.

3.3.4 Primer and paint.

3.3.4.1 Metal,primer. Primer used on metal surfaces to be painted shall conform to TT-P-1757.

3.3.4.2 Finish paint. The finish coat of paint for ferrous and aluminum metal shall conform to type A, table II of MIL-T-704. The finish coat paint shall be in accordance with MIL-E-52798 (forest green).

3.3.5 Aluminum alloy.

3.3.5.1 Sheet or strip. The aluminum alloy sheet or strip shall be 5052, H32 or H34 conforming to QQ-A-250/8.

3.4 Design. The container shall have nominal overall dimensions of 8 feet in width, 8 feet in height and 10 feet in length. The container walls, ceiling, and floor except over fork lift pockets, shall be insulated with a minimum of 3-1/2 inches of insulation specified in 3.3.2. The insulated end wall opposite the door end wall shall have an opening as shown on drawing 5-13-2853 for mounting a 4,000 B.T.U./hour refrigeration unit. The unit opening end wall shall be located 18 inches from the outside of the corner fittings. All four sides and the extreme front of the container near the refrigeration unit shall be open to allow maximum air circulation. Gussets may be installed on the front part of the container frame. When gussets are used they shall not extend more than 18 inches from the outside of the corner posts as shown on drawing No. 5-13-2854. The maximum height of the top front container end frame shall be no more than 4 inches. A 1/2 inch plywood cover shall be provided for the refrigeration unit opening. No part of the container shall protrude beyond the outside surfaces of the corner fittings. The nuts to receive the refrigeration unit shall be size 3/4-10, square, hardened, captive and floating. Nuts shall be treated to resist corrosion. The container end wall with unit opening shall be of sufficient strength to accept the 4,000 B.T.U./hour refrigeration unit having a weight of 500 pounds. Dissimilar metals as defined in MIL-T-704 shall not be used together unless suitably protected against electrolytic corrosion.

3.4.1 Insulation installation. Prior to the insulation application, the sides of the interior and exterior skins exposed to the insulation shall be treated in accordance with the manufacturer's standard procedure to assure permanent adherence of the insulation to the skin material.

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3.4.2 Weight and dimensions. The weight of the container without the refrigeration unit shall not exceed 3500 pounds. Dimensions, tolerances and diagonal differences of the container shall meet the requirements as specified in ANSI MH 5.1.1M, except as specified in table I. All inside dimensions shall be the maximum attainable, consistent with design and material, but not less than those specified in table I.

TABLE I. Inside dimensions

Width, minimum	86 inches
Length, minimum	88 inches
Height, minimum	83 inches

3.4.2.1 CSC approval. In accordance with Title 49 CFR parts 450-453, approval of the International Convention for Safe Containers (CSC) shall be obtained (see 4.4.1.3).

3.4.3 Markings. Containers shall be marked as specified (see 6.2).

3.4.4 Weight markings. The tare weight to be marked on the container shall be the weight of the container plus 500 pounds for the weight of the 4000 ETU refrigeration unit. A gross weight limit of 10,000 pounds shall be marked on the container in white letters not less than two inches in height.

3.5 Construction.

3.5.1 Interior walls and ceiling. Each interior end wall, sidewall, and ceiling shall be of one piece flat aluminum specified in 3.3.5.1 and shall have a minimum thickness of 0.050 inches. Provisions shall be made for air flow in the vertical direction on the interior side and end walls by vertical ribs having a minimum depth of 5/8 inch and spaced 12 inches on center. Interior seams between metal skins shall not have any open crevices.

3.5.2 Exterior walls. The exterior skin material shall be flat aluminum specified in 3,3.5.1 and shall have a minimum thickness of 0.050 inches.

3.5.3 Floor construction. The interior floor shall be of aluminum specified in 3.3.5.1 having a minimum thickness of 0.125 inches and shall be of a pan design, welded to eliminate seepage of moisture into the insulation when tested as specified in 4.5.2.11. The floor shall have a corrugated configuration to provide air flow in the longitudinal direction with the corrugated depth being a minimum of 1-1/4 inches. The floor shall have two drains located at the two comers at the refrigeration unit end. The drains shall be protected by fittings which will open automatically when subjected to pressures above normal internal pressure. Drains shall be provided with manual closures. The top of the floor drains shall be not less than 1/4 inch below the bottom surface of the floor pan.

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3.5.4 Understructure. The container bottom side rails, bottom frame members, and cross members shall not deflect below the bottom plane of the bottom corner fittings when loaded to a uniformly distributed load of 18,000 pounds gross weight as specified in 4.5.2.9.

3.5.5 Roof construction. The roof sheet and upper end frame members shall be at least 1/4 inch below the top plane of the top corner fittings. The roof shall be one piece flat aluminum specified in 3.3.5.1 and installed in a manner to prevent infiltration of water into the insulation area.

3.5.6 Door. The door shall be of the infitting type having the exterior door skin flush with the exterior wall and shall be designed to withstand rough handling. The door jamb shall be constructed with the use of low heat conductive material. The door shall be provided with a bumper to prevent door hardware from damaging exterior container skin when opened. The door shall have a heavy duty type gasket suitable to meet all of the specification requirements. The clear door opening shall be 74 inches in height and 36 inches in width. The door sill shall be of aluminum material and attached to the floor pan by welding and shall not be higher the lowest part of the corrugated floor. The door shall have a heavy duty, handle operated, cam locking device and shall be provided with an internal release handle, The latch construction shall permit a person trapped in the container to escape when the container is padlocked on the outside. The door latch shall have provisions to accept a padlock as specified in 3.13. With the padlock in place it shall prevent the door from being opened from the outside. The door shall be provided with three heavy duty steel hinges and shall have corrosion resistant steel pins. Lubrication fittings shall be installed at each door hinge pin location. No external door hardware shall protrude beyond the container corner fittings. A rain hood shall be provided over the door,

3.5.7 Corner fittings. The corner fittings shall conform **to** dimension requirements of ANSI MH 5.1.1M. The corner fittings shall be constructed of such materials as to enable them to pass the operation and testing requirements as specified herein.

3.5.8 Spacer blocks. Unless otherwise specified (see 6.2), the supplier shall provide two spacer blocks with each container. The spacer block shall provide a 3-inch plus O minus 1/16 inch spacing between the containers when affixed to the top corner fittings. The material and design of the spacer blocks shall be of such as to enable them to pass the operating and testing requirements specified herein. A storage bracket for the spacer blocks shall be provided in the exterior refrigeration unit area of the container,

3.5.9 Fork lift pockets. Two fork lift pockets 34 inches on center shall be provided in accordance with fork pocket requirements of ANSI MH 5.1.lM, The pockets shall be closed on the bottom.

3.5.10 Tiedowns. The two interior side walls shall each have two recessed cargo tie down rings located as shown on drawing No. 5-13-2854.

3.5.11 Shelving. Four individual sets of shelving in accordance with MIL-S-40144 shall be provided. Shelving shall be type I, style 1, size 2, except the length shall be 42 inches in lieu of 48 inches. Provisions shall be made to allow the shelving to stand firmly on the corrugated floor by means of foot plates or other suitable methods. Post clamps shall be provided to join the two center posts together when the individual shelving units are assembled.

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3.6 Performance. The container shall meet the following requirements when tested as specified In 4.5.

3.6.1 Stacking. The container, loaded to a gross weight of 1.8 times its rated maximum gross weight (18,000 pounds) shall withstand, without damage or permanent deformation, the loads imposed on its four corner posts when five standard 20-foot containers, each loaded to 1.8 times its maximum gross weight of 44,800 pounds, are stacked on top of two 10 ft containers coupled together. The containers shall be stacked within overall limits of eccentricity of 1 inch laterally and 1-1/2 inch longitudinally (see 4.5.2.1).

3.6.2 Lifting from top. The container, uniformly loaded to a gross weight of 20,000 pounds, shall be capable of being lifted vertically from the top four corner fittings without damage or permanent deformation (see 4.5.2.2).

3.6.3 Lifting from bottom. The container, loaded as specified in 3.6.2 shall be capable of being lifted from the bottom four corner fittings without damage or permanent deformation (see 4.5.2.3).

3.6.4 Restraint. The container, loaded to its rated maximum gross weight of 10,000 pounds, shall withstand without damage or permanent deformation, 2.0 g's external restraint applied in a horizontal plane in the longitudinal direction (see 4.5.2.4).

3.6.5 Floor. The floor of the container shall withstand without damage or permanent deformation a load of not less than twice the maximum payload when 60 percent of the payload is uniformly distributed over the longitudinal center 5 feet of the floor with the balance of the load uniformly distributed over the remainder of the floor (see 4.5.2.8).

3.6.6 Roof. The container roof shall withstand without damage or permanent deformation a load of 400 pounds uniformly distributed over an area 12 by 24 inches (see 4.5.2.7).

3.6.7 Walls. Each end wall of the container shall withstand, without damage or permanent deformation, a uniformly distributed load of not less than 0.4 times the maximum payload of a 10-foot container (see 4.5.2.5). Each side wall shall withstand, without damage or permanent deformation, a uniformly distributed load of 0.6 times the maximum payload for a 10-foot container (see 4.5.2.6).

3.6.8 Racking, The container shall withstand, without damage or permanent deformation, a lateral and longitudinal concentrated load of 33,600 pounds for 5 minutes to each of the four top corner fittings, when the bottom comer fittings are firmly anchored (see 4.5.2.10).

3.6.9 Weatherproofness. The container door and all joints shall show no signs of leakage when tested as specified in 4.5.2.9.

3.6.10 Heat transfer. The heat transfer of the container shall not exceed 30 B.T.U. per hour per degree F with a mean wall temperature of 50°F; or 33 B.T.U. per hour per degree F with a wall temperature of 100°F when tested as specified in 4.5.2.14.

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3.6.11 Air leakage. The air leakage of the container shall not exceed 20 cubic feet per minute at 1 inch water column when tested as specified in 4.5.2.12.

3.6.12 Container coupling. Two 10-foot containers each with an evenly distributed weight of 10,000 pounds gross shall be coupled together and lifted by the outer corner fittings without damage or permanent deformation (see 4.5.2.13).

3.7 Wire. Electric wire shall be non-metallic sheated cable conforming to type UMC or UF of the National Electric Code. Cable shall contain two conductors and a ground wire. The conductor size shall be No. 14 AWG.

3.8 Switch. The switch for the interior light shall conform to class I, type 1 of W-S-896/1. The switch shall be housed in a vapor-tight inclosure recessed. within the container wall and shall be externally operated. The switch shall be located on the exterior of the container adjacent to the door latch as shown on drawing 5-13-2854.

3.9 Pilot light. A pilot light shall be furnished and shall be enclosed in a vapor-tight housing. A red indicator shall illuminate when the switch is on the “On” position.

3.10 Lighting fixture. The lighting fixture shall be a vaporproof-type fixture and shall be center mounted over the door on the interior of the container. The vaporproof fixture shall be equipped with a shatterproof globe of the polycarbonate screw-type and shall be complete with a 100 watt, 120-volt inside frosted incandescent bulb having a medium screw base.

3.11 Plug and receptacle. A plug receptacle wired to the interior light fixture, switch, and pilot light shall be provided and shall be of the weather-tight type, designed for 3-wire, 2-pole grounded service and shall be rated at 15 amperes, 125 volts. The wiring shall be installed internally within the ceiling structure. The receptacle shall be of the recessed, male, locking type. A cap or flap cover shall be provided to seal the receptacle housing when not in use and shall be secured to the housing to prevent loss. A plug shall be provided and shall contain a sealing ring to form a weather-tight joint when the plug is inserted in the receptacle. The receptacle shall be located on the exterior at the refrigeration unit end of the container on the left side facing the refrigeration unit as shown on drawing No. 5-13-2854.

3.12 Thermometer. A thermometer shall be provided and mounted with the dial face on the exterior front of the container adjacent to the walk-in door. The sensing element shall be mounted to a bracket on the interior of the container midway from floor to ceiling and shall be protected from damage by a guard or other suitable protective device. The range of temperature readings will be -25 to +120 degrees F.

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3.13 Padlock. A padlock conforming to MS 35647-5 shall be provided. The padlock shall be securely attached to the walk-in door by a chain.

3.14 Finish. Unless otherwise specified (see 6.2) all external surfaces shall be finished.

3.14.1 Cleaning, conditioning and surface preparation. All exterior surfaces shall be thoroughly cleaned, conditioned and prepared for painting with compound specified in 3.3.3.1 or 3.3.3.2.

3.14.2 Priming. All metal surfaces prepared for painting shall receive a coat of primer specified in 3.3.4.1.

3.14.3 Finish coat. All surfaces to be painted shall be thoroughly coated with semigloss enamel specified in 3.3.4.2. No paint s hall be used in the interior of the container.

3.14,4 Undercoating. The underside of the container shall be coated after painting with a l/16-inch bituminous coating conforming to TT-C-520.

3.15 Data plate. A US data plate shall be securely attached to the door. Plates will be furnished upon request to the Commandant of the Marine Corps, Code LMO-3, Washington, DC 203800 The contractor shall imprint the data required in all blank spaces.

3.16 Metal fabrication,

3.16.1 Sheet metal. Metal used in the fabrication of the container shall be free of kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major character shall be made with dies or fixtures to insure uniformity of size and shape.

3.16.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

3.16.3 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or defects which might render the castings unsound for use.

3.16.4 Bolted and riveted connectors. Bolt and rivet holes shall be accurately punched or drilled and shall have the burrs removed. Washers, lock-washers, or lock nuts shall be provided where necessary and all bolts, nuts and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter of rivet, concentric with the rivet holes, and in full contact with the surface of the members.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b, Quality conformance inspection (see 4.4) .

4.3 First article inspection.

4.3.1 in-process inspection, As part of the first article inspection, the first two containers produced shall receive an in-process inspection by the contractor under the direction and surveillance of the Government. This inspection shall be conducted to evaluate materials and workmanship as specified in 3.3 and 3.16. Non-conformance to 3.3 or 3.16 s'nail be cause for rejection.

4.3.2 Examination. AS part of the first article inspection; the two containers shall be examined. as specified in 4.4.2.1 Presence of one or more major defects in either container shall be cause for rejection of both containers.

4.3.3 Tests , As part of the first article inspection, one container shall be tested as specified in 4.5.2..1 thru 4.5.2.14. Failure on any test shall be cause for rejection of the container.

4.4 Quality conformance inspection. Unless otherwise specified, sampling and inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1., the components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification.

4.4.1.1 In-process inspection. In accordance with 4.1, the processes used to prepare skin surfaces for insulation adhesion, undercoating, or painting shall be observed to determine conformance to the manufacturer's standard procedures. Any material prepared in non-conformance will not be considered for acceptance.

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4.4.1.2 Certification. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the electrical equipment he proposes to furnish under this specification meets the applicable requirements of UL standards. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the electrical equipment has been tested and conforms to the requirements of the applicable UL standards. The contractor shall also provide satisfactory evidence that the insulation he proposes to furnish under this specification meets the specified fire-retardant and self-extinguishing requirements.

4.4.1.3 Approval plate. An attached safety approval plate conforming to part. 451 subpart C of Title 49, Code of Federal Regulations shall be presumptive evidence that the container has been approved by an Approval Authority.

4.4.2 End item inspection.

4.4.2.1 End item examination.

4.4.2.1.1 Visual examination. Examination shall be made of each container for the defects listed in table II, and any major defect shall be cause for rejection of the unit.

TABLE 11. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished where required	X	
	Color not as specified		X
	Component or assembly painted or coated where not required	X	
Design	Not design specified	X	
Plywood cover	Not in place	X	
Installation of insulation	Skins not bonded	X	
Safety plate	Not installed	X	
Weight markings	Not legible		
Interior walls and ceiling	Not design specified	X	
Floor construction	Not design specified	X	
	Not welded as specified	X	

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TABLE II. Classification of defects (cent'd)

Examine	Defect	Classification Major	Minor
Floor drains	Missing, not as specified	x	
Door latch	Not design specified, internal release missing	x	
	Not design specified	X	
Spacer blocks (when specified)	Missing, not number specified	x	
Storage bracket	Missing, not as specified	x	
Tiedowns	Missing, not located where specified	x	
Wiring	Not as specified	x	
Switch	Missing, not design specified, not located where specified	x	
Pilot light	Missing, not design specified, not color specified		X
Lighting fixture	Missing, not design specified	x	
	Not located where specified		x
Plug & receptacle	Missing, not design specified	x	
	Not located where specified		x
Padlock	Missing, not design specified	x	
	Not attached as specified		x
Data plate	Missing	x	

4.4.2.1.2 Dimensional and weight examination. Examination shall be made of each container unit for defects in dimensions or weight. Any dimension or weight, not within specified requirements, shall be cause for rejection of the unit.

4.4.2.2 End item testing. Each container shall be tested as specified in 4.5.2.9. One out of 25 containers shall be tested as specified in 4.5.2.12 and 4.5.2.13. Failure to pass any test shall constitute rejection of the container and the lot it represents.

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4.4.3 Packaging inspection. An examination shall be made to determine that preservation, packing and marking as required by section 5 are complied with. Defects shall be scored as specified in table III. The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one container fully packaged. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units.

TABLE III. Packaging inspection

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components, such as incomplete taping or plywood not flush against refrigerator.

4.5 Test methods.

4.5.1 Test conditions. The container shall first be tested for weather-proofness before application of finish painting of undercoating. Other tests shall be made after application of finish painting and undercoating. A second test for weatherproofness shall be made after completion of all other tests. The test load within the container shall be uniformly distributed, except as othewise specified herein. On completion of each test, the container shall be examined to determine compliance with the applicable performance requirement specified in 3.6. During tests, tare weight of container is the empty weight of container.

4.5.2 Test procedures.

4.5.2.1 Stacking. Two 10 ft containers coupled together shall be placed on four level pads, one under each of the four external end bottom corner fittings. The pads shall be centered under the fittings and shall be the same plan dimensions as the fittings. The containers shall each be uniformly loaded to a gross weight of 18,000 pounds. Five standard 20-foot containers, each uniformly loaded to a gross weight of 80,640 pounds, shall be stacked on top of the containers under test offset by 1 inch laterally and 1-1/2 inch longitudinally, The five containers shall remain in place not less than 5 minutes. The test shall then be repeated with the five containers offset longitudinally in the opposite direction.

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4.5.2.1.1 Alternate stacking. In lieu of the procedure outlined in 4.5.2.1, the container under test, loaded to a gross weight of 18,000 pounds and placed as specified in 4,5.2.1, shall be subjected to a load of 403,200 pounds. The load shall be applied through the centers of four pads of the same plan area as the corner fittings and shall be equally divided among the four corner fittings. Each pad shall be offset in the same direction by 1 inch laterally and 1-1/2 inch longitudinally-. The load shall be applied for not less than 5 minutes. The test shall then be repeated with pads offset longitudinally in the opposite direction.

4.5.2.1.2 Alternate individual corner structures. In lieu of the procedure outlined in 4.5.2.1.1, all comer structures may be test loaded individually to 100,800 pounds. The container shall be loaded to a gross with of 18,000 pounds. The test shall be made on a container, placed on pads as specified in 4.5.2.1. The load shall be applied through the center of a rigid pad of the same dimensions as the corner fittings. The pad shall be offset from the corner fitting 1 inch laterally and 1-1/2 inch longitudinally. Direction of offset shall be changed so that in testing two containers each possible offset is applied to the rear and to one front corner fitting. The lot shall be applied for not less than 5 minutes. No side or end supports for the container shall be used in this test.

4.5.2.2 Lifting from the top. The container under test shall be loaded to a grass weight of 20,000 pounds. The container shall be lifted by all four top comer fittings in such a way that no noticeable acceleration or deceleration forces are applied. The lifting force shall be applied vertically. The container shall be lifted until clear of the supporting surface and suspended for not less than 5 minutes. Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.3 Lifting from the bottom. The container under test shall be loaded to a gross weight of 20,000 pounds. The container shall be carefully lifted by all four bottom corner fittings in such a way that no noticeable acceleration or deceleration forces are applied. The lifting force shall be applied using one spreader above the roof. Lifting shall be performed in such a way that the lifting cables do not bear against the container sides. The line of action of the lifting cables and the outer face of the comer fitting shall be parallel and no further apart than 1-1/2 inches. The containers shall be lifted until clear of the supporting surfaces, and suspended for not less than 5 minutes. Nonconformance to 3.6.3 shall constitute failure of this test.

4.5.2.4 Restraint. The container under test, loaded to a gross weight of 10,000 pounds shall be restrained longitudinally by securing the bottom corner fittings at one end to suitable anchor points through the bottom apertures. A force equal to 20,000 pounds shall be applied longitudinally to the container through the end apertures of the bottom corner fittings at the opposite end, first

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in compression and then in tension, for not less than 5 minutes each. A force of 10,000 pounds in tension or compression shall be borne by each lower corner fitting and its connection to the container structure. Nonconformance to 3.6.4 shall constitute failure of this test.

4.5.2.5 End wall strength. A 10-foot container shall be loaded with a weight equal to 0.4 times the maximum container payload (maximum container payload equals 10,000 pounds, less the tare weight of the container including unit) . Water or air shall be used for the test load. The container, with the end to be tested facing down, shall be supported on the level pads placed under the end faces of the two bottom corner fittings. The container shall be held in this position by restraining devices attached to the bottom corner fittings at the opposite end. The supporting pads shall be of sufficient thickness to allow the ends of the container to cantilever freely in space when loaded as specified. Structural members and panels not under test may be supported during this test. The load shall be applied to the full cross-section of the end wall, and shall be uniformly distributed. Tests shall be made on both the front end and on the rear (door) end. The load shall remain in place for not less than 5 minutes. A test on the door and door frame not attached to a container is not an acceptable substitute for the specified test on the door end. The refrigeration unit shall not be installed in the container during wall strength tests. Nonconformance to 3.6.7 shall constitute failure of this test. It is not necessary to stand the container on end when air testing.

4.5.2.6 Side wall strength. The container shall have one side wall tested and shall be loaded to 0.6 times its maximum payload. Water or air shall be used for the test load. The container, with the side to be tested facing down, shall be supported on two level pads placed under the side faces of the two bottom corner fittings. The container shall be held in this position by restraining devices attached to the bottom corner fittings at the opposite side. The supporting pads shall be of sufficient thickness to allow the sides of the container to cantilever freely in space when loaded as specified, The load shall be applied to the full cross-section of the side wall and shall be uniformly distributed. The roof may be supported during this test. The load shall remain in place for not less than 5 minutes. Nonconformance to 3.6.7 shall constitute failure of this test. It is not necessary to stand the container on end when air testing.

4..5.2.7 Roof strength. A load of 400 pounds shall be uniformly distributed. over an area of 24 inches by 12 inches located with the 12-inch dimension parallel to the longitudinal axis of the container, between roof supporting cross-members, and as nearly as possible equally distant between front and rear and between the sides of a 10-foot container. The load shall remain in place for not less than 5 minutes. Nonconformance to 3.6.6 shall constitute failure of this test.

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4.5.2.8 Floor strength. Rest the container on its four bottom comer fittings, with the intermediate structure a minimum of 1 inch above the supporting surface and free to deflect. The container shall then be loaded to not less than two times the maximum single container payload. Sixty percent of the load shall be uniformly distributed over the longitudinal center 5 feet of the floor and the other 40 percent shall be uniformly distributed over the remainder of the floor. Load shall remain in place for not less than 5 minutes. Failure to meet the requirements of 3.5.4 and 3.6.5 shall constitute failure of this test.

4.5.2.9 Weatherproofness. A stream of water at variable angles shall be applied to all joints, seams and gasketed closures except during racking tests of the container from an 0.5 inch inside diameter nozzle at a nozzle pressure of not less than 15 pounds per square inch (psig). Rate of movement over the exterior of the container shall be approximately 4 inches per second. The nozzle shall be held 5 feet away from the container under test. The test shall be continued for not less than 15 minutes. on completion of the test, the container shall be examined for penetration of inter into the interior of the container. Nonconformance to 3.6.9 shall constitute failure of this test.

4.5.2.10 Racking. The container under test, loaded to a gross weight of 10,000 pounds, shall be restrained through the bottom apertures of all four bottom corner fittings. The door shall be closed and latched. A concentrated force of 33,600 pounds shall be applied first laterally and then longitudinally to each top corner fitting in turn. Each force application shall be not less than 5 minutes. Nonconformance to 3.6.8 shall constitute failure of this test.

4,5.2.11 Floor pan waterproofness. The container floor pan shall be tested for waterproofness, by completely filling the floor pan with water to a depth of not less than 1 inch, Duration of the test shall be not less than 8 hours. Any water leakage as indicated by a reduction in water depth shall constitute nonconformance to 3.5.3 and shall constitute failure of this test.

4.5.2.12 Air leakage. The opening for the refrigeration unit shall be closed with an insulated plug having the same insulation thickness as the container end wall, The container shall have inside and outside temperatures stabilized within the range of 55°F to 77°F. Drain openings are to be closed. An air supply through a metering device and a suitable pressure measuring device (manometer) is to be connected to the container by a leakproof connection. The flow measuring device shall be accurate to + 3 percent of the measured flow rate and the pressure measuring device on the container shall be accurate to + 5 percent of the reading. Air is to be admitted to the container to raise the internal pressure to l.0 + 0.04 inches of water column and the air supply regulated to maintain the pressure for at least 3 minutes. When the steady test conditions have been established for at least 3 minutes, the air flow required to maintain the pressure is to be recorded. The air leakage rate corrected to standard atmosphere conditions of 68°F and 14.7 PSIG shall not exceed 20 cubic feet per minute (CFM) at l-inch water column in compliance with 3.6.11. Leakage rates in excess of 20 CFM shall constitute failure . of this test.

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4.5.2.13 Coupled test. A 20-foot configuration, two 10-foot containers, each loaded to a gross weight of 10,000 pounds, shall be coupled together by couplers between lower corner fittings. Each opening for the refrigeration unit shall be closed with an insulated plug having the same insulation thickness as the container and wall. Spacer blocks shall be used between upper corner fittings of the two containers. The coupled containers shall be lifted a minimum of 1 foot ground level and suspended for a minimum periof of 5 minutes--to determine compliance with 3.6.12.

4.5.2.14 Heat leakage test. The opening for the refrigeration unit shall be closed with an insulated plug having the same insulation thickness of the container end wall. The container shall be placed in a controlled temperature ambient and a heat leakage test conducted to determine compliance with 3.6.10. All calculations using the formula below and all data on time, temperature and heat input shall be recorded. The ambient temperature shall be the average of eight readings taken at points located 6 inches from each outside corner of the container. The internal container temperature shall be the average of eight readings taken at points located 6 inches from the inner walls at each corner of the container. Eight ambient and interior temperature readings shall be taken at consecutive 1/2 hour intervals with ambient temperature variations not more than plus or minus 2°F and interior container variations not more than plus or minus l°F. One interior and one exterior temperature reading shall be taken on a recording thermometer. Sufficient steady heat shall be added to the container to maintain a minimum of 60°F and a maximum of 100°E temperature difference between *the* interior and exterior of the container. The interior of the container shall not exceed 1550??. An electric fan shall be placed within the interior of the container to provide interior air circulation. All electrical heat input including electrical energy from fan motor shall be measured by a watt-hour meter. The heat leakage of the container shall be determined as follows:

$$\text{Heat Leakage (B.T.U./hr/°F)} = \frac{\text{Watt hrs. input X B.T.U./watt hr}}{\text{Length of test (hrs) X temperature Difference (°F)}}$$

Heat transfer rates in excess of those specified in 3.6.10 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Refrigeration unit opening. The plywood cover plate specified in 3.4 shall be securely installed over the refrigeration unit opening.

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5.1.1.2 Thermometer. The thermometer shall be covered and protected with plywood fitted to bear flush against the refrigerator. The plywood shall be secured in place by means of strips of tape covering all sides and extending onto the surface of the refrigerator. The tape shall be minimum 2-inch width conforming to type III or IV, class 1 of PPP-T-60. The plywood shall be not more than l/4-inch thick and shall conform to NN-P-530.

5.1.1.3 Spacer blocks and electrical plug. The spacer blocks and electrical plug shall be unit packed for mechanical and physical protection within a snug-fitting fiberboard box conforming to style RSC, class domestic of PPP-B-636. The box shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.1.1.4 Shelving. Each set of shelving shall be unit packed as specified in MIL-S-40144.

5.2 Packing. Packing shall be level A.

5.2.1 Level A. Four sets of shelving and spacer blocks and plug, preserved as specified in 5.1, shall be secured within the refrigerator in such a manner as to prevent movement or damage while in transit. The door shall be closed and locked with the padlock. The keys shall be taped to the padlock with tape specified in 5.1.1.2.

5.3 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The container is intended for Marine Corps use in the transportation and storage of food and other temperature sensitive items requiring controlled temperature.

6.2 Ordering data. Procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is not required (see 3.1, 4.3 and 6.3).
(c) When spacer blocks are not required (see 3.5 .8) .
(d) When external surfaces are not be finished (see 3.14).
(e) Markings required.

6.3 'First article. The first article shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services procurement Regulation Tine first article should be the first two containers produced. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

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6.4 Terms and definitions. The terms and definitions used in-this specification shall be as defined in ANSI MH 5.1.1.M.

6.5 Recycled materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.6 Supersession data. This specification includes the requirements of LP/DES S-9-7 dated 16 December 1977.

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Review activities:

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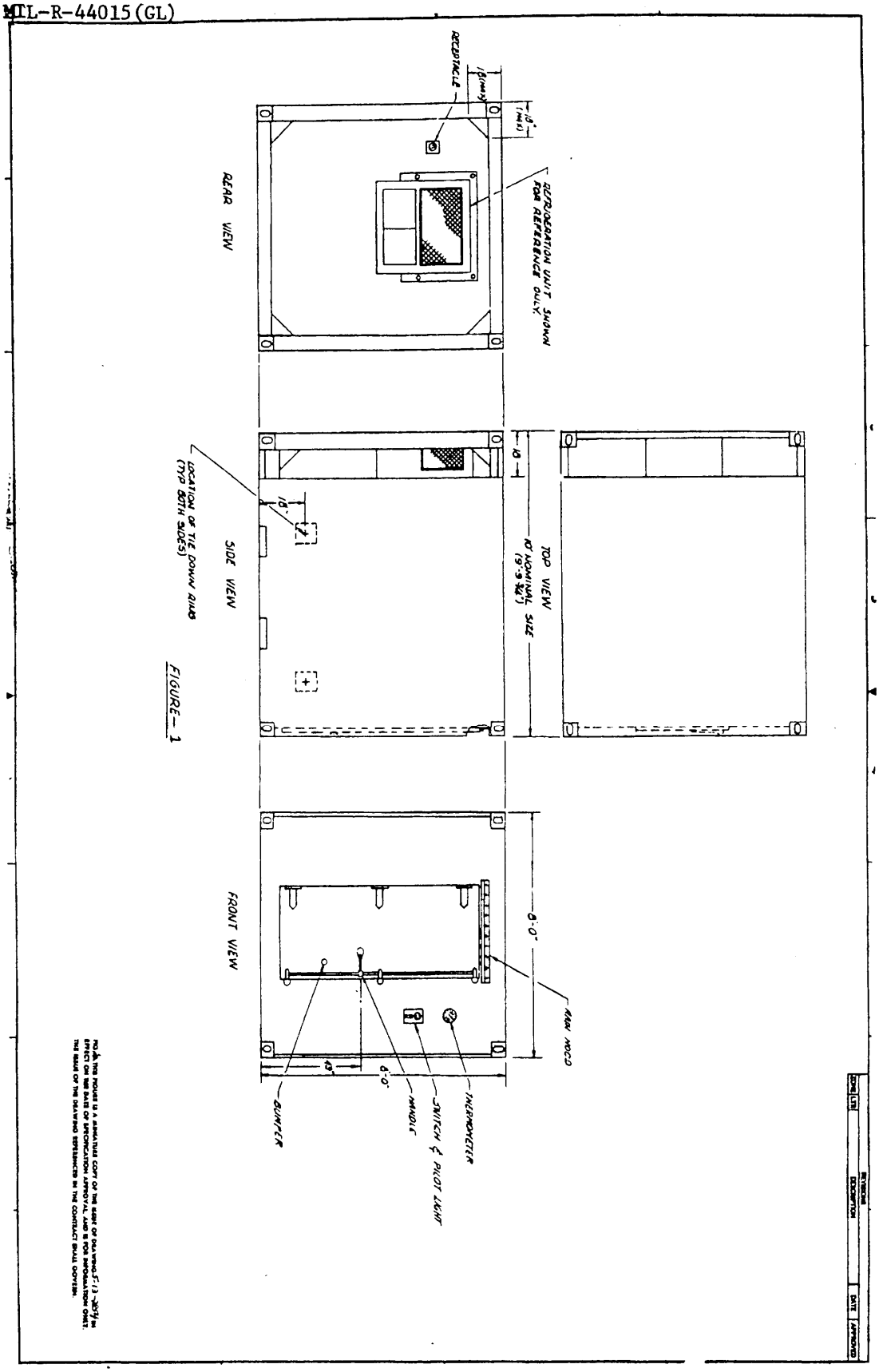


FIGURE 1

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-R-44015(GL)
REFRIGERATOR-CONTAINER, FIELD, 8 FEET BY 8 FEET BY 10 FEET

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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